Lecture VI: Computational Kindness

Programming: Everyday Decision-Making Algorithms

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Today in one sentence

The best algorithms don't just solve a problem, they make life easier and fairer for the people affected.

Why it matters

"When works for you?" sounds polite but hands the work to the other person. Proposing two concrete slots is kinder. Companies face the same question at scale: how to design processes, products and automated decisions that are easy and fair for customers and colleagues.

Roadmap

1. Looking back: five ideas for better decisions
2. Computational kindness: easy and fair for others
3. Looking ahead: where to go from here

Looking back: five ideas for better decisions

Five ideas from this course

- **Optimal stopping:** when to stop looking
- **Explore and exploit:** balance the new and the known
- **Caching:** keep what you use close, let go of the rest
- **Scheduling:** your goal decides your order
- **Randomness:** simulate what you can't calculate

Looking back: five ideas for better decisions

What we covered

Optimal stopping, explore and exploit, caching, scheduling and randomness give you a toolkit for everyday decisions.

Computational kindness: easy and fair for others

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Computational Kindness

Question: **An idea what that is?**

- Idea of introducing empathy in algorithms
- **Consider human cognitive load and limitations**
- Making choices that **reduce mental burden** for others
- Creating systems that are **easy to understand and interact**

Propose two slots instead of asking “when are you free?”

- Propose specific times!
- Don’t ask “when are you free?”
- **Reduces** cognitive load for others
- **Transforms** an open-ended problem

> ! Important
>
> **Now a simple yes/no decision!**

Say what you prefer

- Be explicit rather than implicit
- Provide **clear** options instead
- **Avoid** open-ended questions
- State your **preferences**

> ! Important
>
> **Avoid** deferring → “I’m fine with whatever”!

“I’m free whenever” is not helpful

- “I’m free whenever” seems helpful
- But it is not!
- Forces others to **consider all possibilities**
- **Increases** cognitive load
- Makes decision-making **more complex**

Four principles of computational kindness

1. **Reduce Options**: Fewer choices lead to better decisions
2. **Be Explicit**: Clear constraints help others decide

3. **Consider Cognitive Load**: Design interactions minimizing mental effort for others
4. **Make Decisions**: Taking responsibility can be kind

Biased data leads to biased decisions

- Algorithms increasingly decide **about people**: credit, job ads, CV screening
- They learn from **historical data**, including its biases
- Amazon dropped a CV-screening tool that downgraded CVs mentioning "women's"[1]

Kindness also means thinking about the **people affected** by our algorithms.

Where bias comes from

- **Data:** some groups are missing (selection bias, Lecture V)
- **Objective:** optimizing only speed or profit ignores who bears the errors
- **Feedback loops:** decisions shape the next data[2]

Fairness is a design decision: **test** outcomes for groups, **explain** decisions, keep a **human** in the loop.

Three questions before you deploy an algorithm

Question: **Before you deploy an algorithm, ask:**

- Who is affected, and who is **missing** in the data?
- Can a person **understand and contest** the decision?
- Does it **reduce or add burden** for the people involved?

Your turn: be kind in three situations

Think about it: **How could you apply computational kindness here?**

- Planning a **group dinner** with friends
- Asking your professor for **thesis feedback**
- Coordinating a **team project** at work

Computational kindness: easy and fair for others

What we covered

Good algorithms reduce the burden for others and treat people fairly. Biased data leads to biased decisions.

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[1]Dastin, J. (2018). Amazon scraps secret AI recruiting tool that showed bias against women. Reuters, 10 October 2018.

[2]Ensign, D. et al. (2018). Runaway feedback loops in predictive policing. Proceedings of Machine Learning Research, 81, 160–171.

Looking ahead: where to go from here

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How to continue learning?

- We have covered **a lot of topics**
- But there are **many more** to explore!

Bayes' rule: update your beliefs with new evidence

- Fundamental theorem in probability theory
- Updates probability of hypothesis based on new evidence
- Used in **statistics, machine learning, and decision-making**

> ! Important
>
> Helps make **informed predictions and decisions** under uncertainty in the real world!

Overfitting: a model that learned the past too well

- Occurs when a model learns the training data too well
- **Captures noise and outliers** rather than pattern
- Results in **poor generalization to new data**

> ! Important
>
> Several ways to counter overfitting, e.g. cross-validation.

Game theory: decisions when others decide too

- Strategic interactions among rational decision-makers
- Analyzes situations with **multiple agents** and their strategies

> ! Important
>
> **Many applications** in economics, politics, and biology!

Networking: how information travels

- Study of **information exchange** over networks
- Includes understanding protocols and data transmission

- Optimizes **network resources**

> ! Important
>
> **Important** in computer networks, the internet and social life!

Relaxation: solve an easier version first

- Simplifies complex problems to make them more tractable
- Involves **relaxing certain constraints**
- Provides insights or **approximate** solutions

Keep programming

- The best way to keep learning is to keep programming
- In your studies, in your thesis, in your job
- **There are many interesting topics to explore!**

Advent of Code: a puzzle a day in December

- Advent of Code is a fun way to keep programming
- Here you can solve programming puzzles during Advent
- It is **completely free** and ad-free and starts each year at 01.12.

Looking ahead: where to go from here

What we covered

Many more ideas from computer science help in everyday life, and the best way to keep learning is to keep programming.

What we covered

Looking back: five ideas for better decisions

Optimal stopping, explore and exploit, caching, scheduling and randomness give you a toolkit for everyday decisions.

Computational kindness: easy and fair for others

Good algorithms reduce the burden for others and treat people fairly. Biased data leads to biased decisions.

Looking ahead: where to go from here

Many more ideas from computer science help in everyday life, and the best way to keep learning is to keep programming.

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scale: how to design processes, products and automated decisions that are easy and fair for customers and colleagues.

Make it easy for others.

That's it for the lecture series!

- We have covered the basics of Python and six ideas for better decisions
- If you have questions or feedback, please let me know!
- All the best for your studies and your career!

Literature

Interesting literature to start

- Christian, B., & Griffiths, T. (2016). Algorithms to live by: the computer science of human decisions. First international edition. New York, Henry Holt and Company.[3]

Books on Programming

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. Free online
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

> **i** Note
>
> Think Python is free online and a great start. Schrödinger programmiert Python is a playful alternative in German.

More Literature

For more interesting literature, take a look at the literature list of this course.

[3] The main inspiration for this lecture. We have read it and discussed it in depth, always wanting to translate it into a course.